signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

*signal.sci

```
// Test Case: Convolution of Discrete-Time Signals
// Define the discrete time signals
x_n = [1, 2, 3]; // Signal x[n]

h_n = [1, -1]; // Signal h[n]
// Compute the convolution
y_n = conv(x_n, h_n); // Convolution of x[n] and h[n]
// Define the time index for the output signal
n_y = 0:(length(x_n) + length(h_n) - 2); // Length of y[n]
// Plot the input signals and their convolution
clf; // Clear the current figure
// Plot x[n]
subplot(3,1,1);
bar(0:length(x_n)-1, x_n, 'b');
xlabel('n');
ylabel('x[n]');
title('Signal x[n]');
// Plot h[n]
subplot(3,1,2);
bar(0:length(h_n)-1, h_n, 'r');
xlabel('n');
ylabel('h[n]');
title('Signal h[n]');
// Plot y[n] (convolution result)
subplot(3,1,3);
bar(n_y, y_n, 'g');
xlabel('n');
ylabel('y[n]');
title('Convolution of x[n] and h[n]');

```

Line 31, Column 0.

